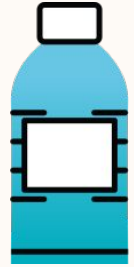
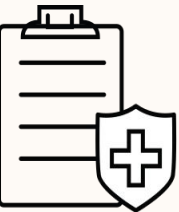


Pharmacy Database Managment System



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Contents

01 Project Overview

02 Connection to Data Science

03 Project Goals & Goals Achieved

04 Database Design

05 Database Creation & Features Implemented

06 Business Value

07 Demo & Future Improvements



What the Project is About

- The purpose of the system is to organize and manage important pharmacy data in a structured way.
- Store information on drugs, customers, prescription, employees, purchases, sales, and inventory.
- Use normalization to reduce and create related tables.
- Support efficient data retrieval and further analysis.



How this Project Connects to Data Science



DATA ORGANIZATION

- Stores patient, prescription, medication, and inventory data.
- Structures data into tables for easy access and management
- Ensures data is clean, consistent, and usable.



PATTERN & TREND ANALYSIS

- Tracks prescription frequency and refill patterns.
- Monitors inventory levels and medication demand.
- Helps identify trends to support better decision-making.



EFFICIENCY & ACCURACY

- Reduces errors through organized and centralized data.
- Improves speed of accessing patient and prescription info.
- Supports better outcomes through reliable data.

Database Design

To design a normalized pharmacy database to store drugs, customers, prescriptions, purchases, sales, employees, and inventory in a structured way.

Data Import

To convert raw product data into a structure format suitable for the final database design.

Normalization

To reduce redundancy by separating data into related tables and linking them with primary keys and foreign keys.



Project Goals & Goals Achieved

SQL Features

Created SQL objects including a view, function, procedure, and trigger to support reporting and automation.

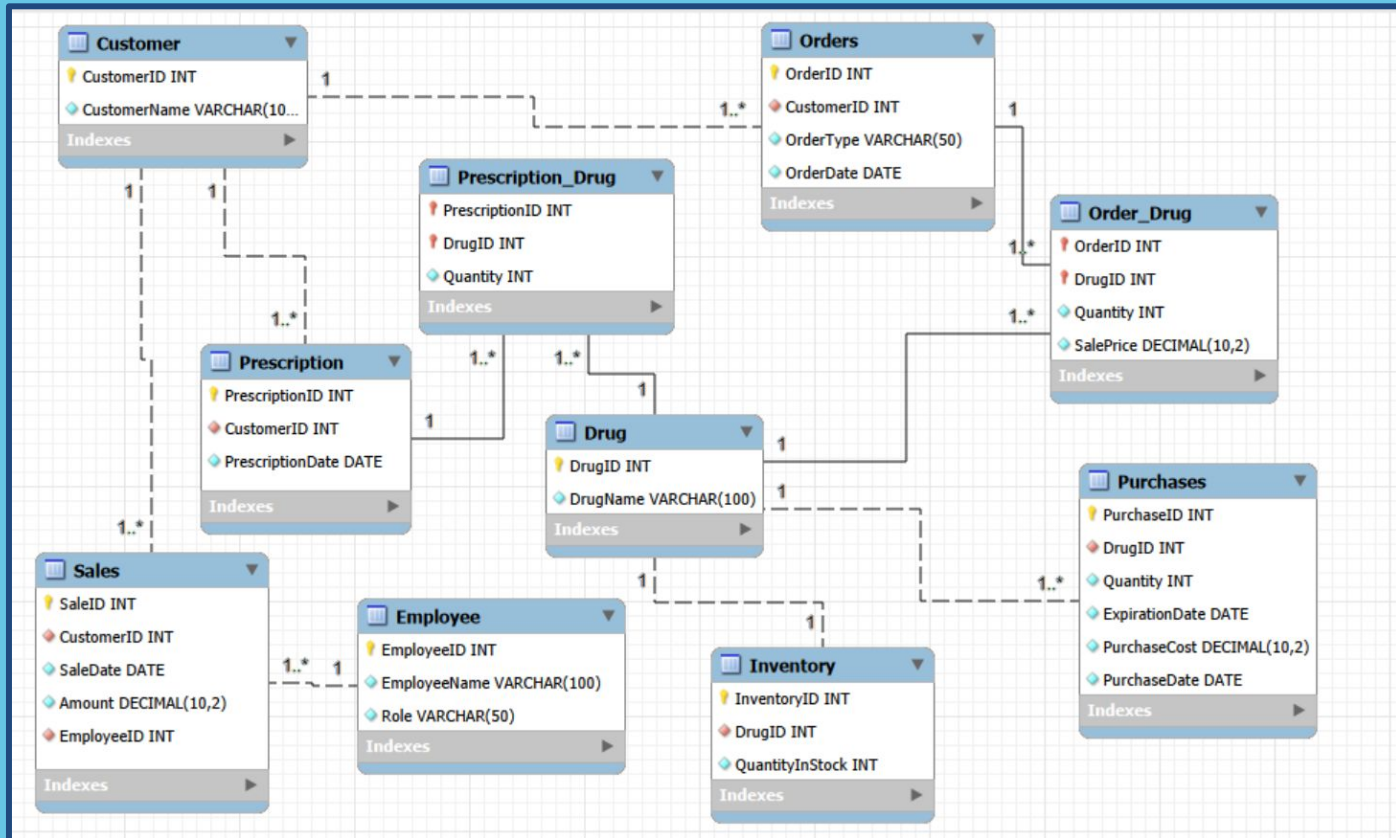
System Functionality

Built a working database system that supports inventory, sales records, purchase records, and prescription management.

Project Outcome

Successfully created a pharmacy database that improves data organization, supports accurate retrieval, and prepare the data for future analysis.

Database Design



Database Creation & Features

Implemented

- Built normalized tables for drugs, customers, employees, prescriptions, purchases, sales, and inventory.
- Imported clean product data from CSV and inserted cleaned data into the final tables.
- Implemented a view to calculate margins.
- Added a function to check product prices..
- Created a procedure to add purchase records.
- Used a trigger to automatically update inventory.

```
-- -----  
-- Table `Pharmacydb`.`Drug`  
-- -----  
  
CREATE TABLE IF NOT EXISTS `Pharmacydb`.`Drug` (  
  `DrugID` INT NOT NULL AUTO_INCREMENT,  
  `DrugName` VARCHAR(100) NOT NULL,  
  PRIMARY KEY (`DrugID`))  
ENGINE = InnoDB;
```

	InventoryID	DrugName	QuantityInSto...
	2	Maalox Antacid Oral Suspension Liquid Lemon...	220
	10	Disposable 5ml Syringe - latex free	100
	11	Panadol 500mg Acetaminophen for Acute Head...	100
	12	Elica-M 0.1% Mometasone Furoate & 2% Micon...	100
	13	Macro Panthenol Softening, Repairing & Protect...	100
	14	Triactin 4mg Cyproheptadine HCL	100
	15	Ambezim-G 5mg Trypsin & 5mg Chymotrypsin	100
	16	Ganaton 50mg Itopride Hydrochloride	100
	17	Lamifen 1% Terbinafine Hydrochloride Anti-Fun...	100
	18	Nexium 20mg Esomeprazole	100
	19	Rowachol Medication for Hepatic, Digestive & G...	100

Business Value of the System

Faster Service Delivery

- Provides quick access to patient and prescription information.
- Improves customer service by reducing wait times.
- Ensures accurate medication dispensing.

Centralized Data Management

- Stores all pharmacy data in one organized system.
- Supports real-time updates for prescriptions and inventory.
- Allows easy access and retrieval of information

Improved Patient Care

- Helps pharmacists make informed decisions.
- Reduces errors through structured and reliable data
- Improves overall quality of patient care

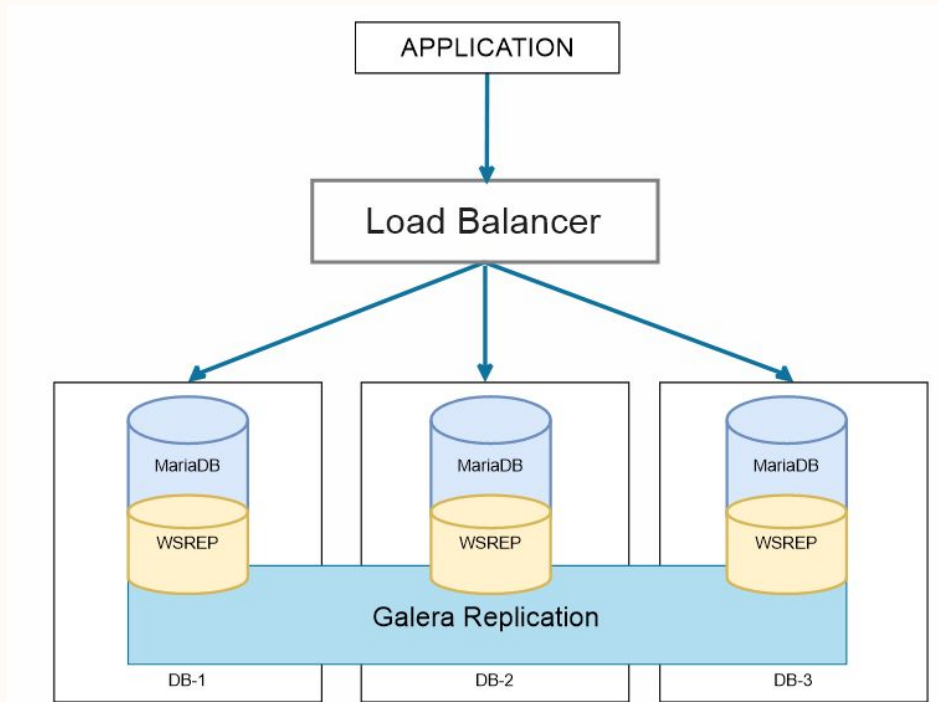


Database Technology Stack: MariaDB with Galera Cluster

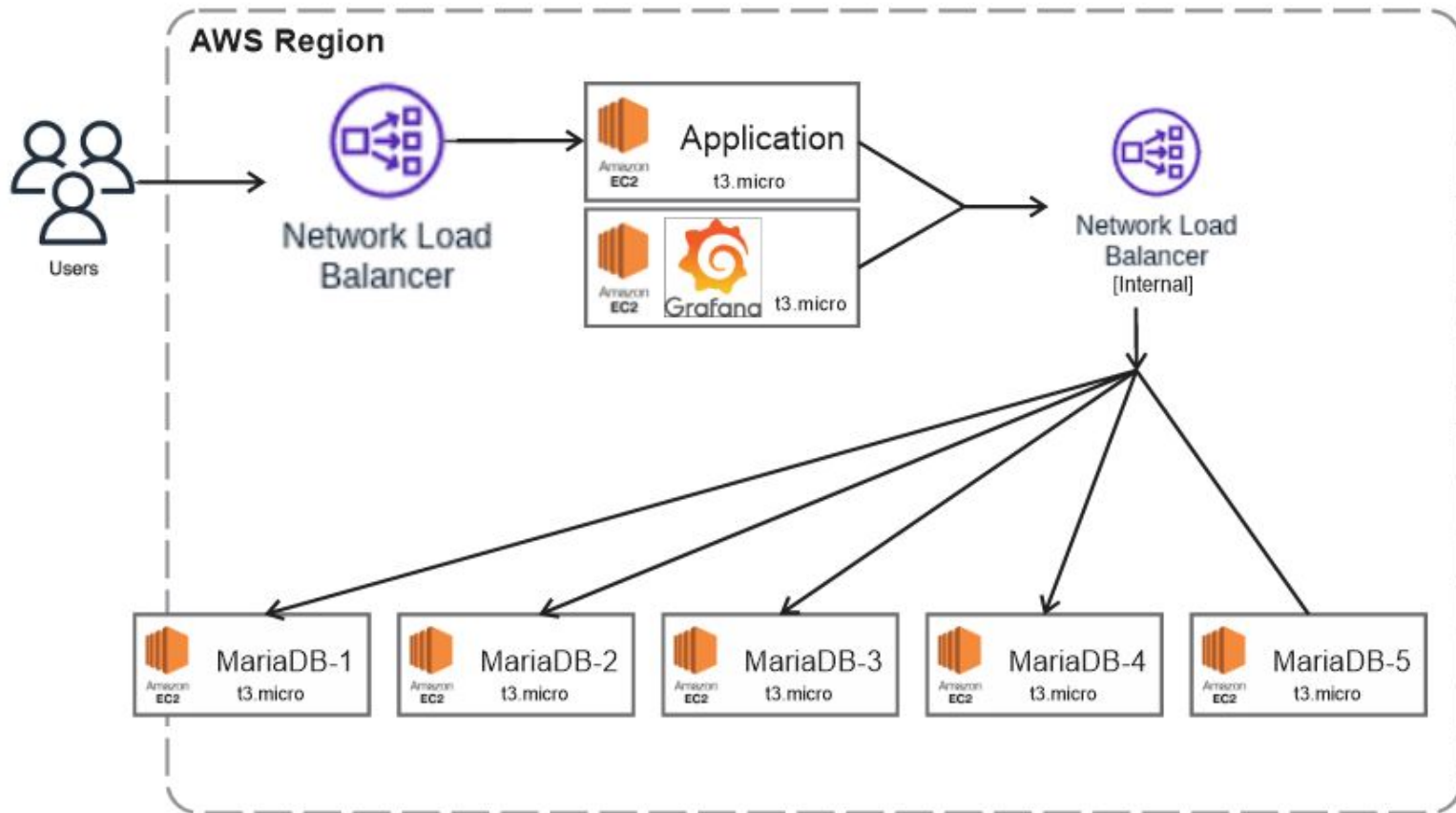
- MariaDB: open source; community developed fork of MySQL
 - Uses SQL for data storage and querying
 - High performance, scalable, and widely used in production systems
- Galera 4: Synchronous multi-master clustering solution for MariaDB
 - Allows multiple nodes to accept reads & writes
 - Uses certification-based replication:
 - Transaction is checked for conflicts before commit on all nodes
 - Provides synchronous replication - no lag like async replication

Database Technology Stack: MariaDB with Galera Cluster

- Galera and CAP Theorem
 - Consistency
 - Partition Tolerance
 - No availability → stop writes if quorum is lost



AWS Deployment Diagram





Demo:

→ <http://nlb-1-ae34f2b35a80be72.elb.us-east-1.amazonaws.com/public-dashboards/c25e2a2c833d4cedb228ad0002534253?from=now-6M&to=now&timezone=browser&refresh=1m>
→ <http://nlb-1-ae34f2b35a80be72.elb.us-east-1.amazonaws.com:8080/admin/>



Conclusion & Future Improvements

PROJECT IMPACT

- Improved organization of pharmacy data through a structured relational database.
- Created a system that supports inventory tracking, sales purchases, and prescriptions.



TECHNICAL ACHIEVEMENTS

- Integrated database with Grafana for analytics purposes.
- Built dashboards to monitor: Sales trends, Orders per day



NEXT STEPS

- Centralize user provisioning, authentication, and authorization.
- Enable data-driven insights by implementing pipelines using Apache Airflow.





Thank you! Any Questions?

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